Comptia N+ Assignment

[N+ - Network security, Maintenance and Troubleshooting procedures]

Network Threats and Mitigation & Physical Security and Risk

Task:

❖ **1. Create a table listing 5 common network threats (such as phishing, DDoS, man-in-the-middle, malware, and password attacks), and for each, write one real-world example from an app or service you use (like WhatsApp, Instagram, or Paytm).**

➢ **Aim**

- To identify five common network threats and understand their real-world impact on applications and online services such as WhatsApp, Instagram, and Paytm.

➢ **Introduction**

- A **network threat** is any activity or attack that can harm a computer network, steal sensitive information, interrupt services, or provide unauthorized access to a system.
- Applications such as WhatsApp, Instagram, and Paytm handle important user information, including messages, account details, and payment-

related data. Therefore, protecting these applications from network threats is essential.

- The following table describes five common network threats with real-world examples.

➢ Common Network Threats and Real-World Examples

No.	Network Threat	Description	Real-World Example
1	Phishing	Phishing is a technique in which attackers send fake messages, emails, or links to trick users into sharing sensitive information, such as passwords or OTPs.	A user receives a fake Instagram message claiming that their account will be suspended. The message contains a link to a fake login page designed to steal their username and password.
2	DDoS (Distributed Denial of Service)	A DDoS attack occurs when many compromised devices send excessive traffic to a server, making the service slow or unavailable to genuine users.	Imagine an online payment service such as Paytm receiving an unusually large amount of malicious traffic. Genuine users may experience delays or difficulty accessing the service.
3	Man-in-the-Middle (MITM)	In a Man-in-the-Middle attack, an attacker secretly intercepts or interferes with communication between two parties.	A user connects to an unsafe public Wi-Fi network while accessing Instagram. If the connection is not properly protected, an attacker may attempt to intercept or

			manipulate network communication.
4	Malware	Malware is malicious software designed to damage devices, steal information, or gain unauthorized access to systems.	A WhatsApp user receives a suspicious file or link from an unknown sender. If the user installs a malicious application through it, the malware may compromise information stored on the device.
5	Password Attacks	Password attacks involve attempts to guess, steal, or obtain a user's password to gain unauthorized access to an account.	An attacker repeatedly tries common or previously leaked passwords to access an Instagram account. A weak or reused password can make the account more vulnerable.

- **Prevention and Mitigation Measures**
- The following security measures can help protect users and applications from these threats:
  1) **Phishing:** Avoid clicking suspicious links and never share passwords or OTPs with unknown people.
  2) **DDoS:** Use firewalls, traffic monitoring, and DDoS protection services to identify and manage abnormal traffic.
  3) **Man-in-the-Middle:** Use trusted networks and secure HTTPS connections. Avoid performing sensitive activities on untrusted public Wi-Fi.

4) **Malware:** Install applications only from trusted sources and keep antivirus software and operating systems updated.
5) **Password Attacks:** Use strong and unique passwords, enable two-factor authentication (2FA), and avoid reusing passwords.

➢ **Result**

➢ Five common network threats—Phishing, DDoS, Man-in-the-Middle, Malware, and Password Attacks—were identified and studied with examples related to commonly used applications and online services.

❖ **2. Simulate a brute-force attack scenario by writing a simple Python script that tries to guess a 4-digit PIN code by checking all possible combinations against a preset value. The script should print the number of attempts it took to guess the correct PIN.**

➢ **Aim**

- To create a simple Python program that simulates a brute-force attack by checking all possible 4-digit PIN combinations against a preset PIN and displaying the number of attempts required to find the correct PIN.

➢ **Introduction**

- A **Brute-Force Attack** is a method in which different possible password or PIN combinations are systematically checked until the correct one is found.
- In this practical, Python is used to demonstrate a basic brute-force simulation in an offline environment. The program checks 4-digit PIN combinations, starting from 0000, until it matches the preset PIN.

- A 4-digit PIN has a total of 10,000 possible combinations, ranging from 0000 to 9999.

➢ 3. Python Program

```
# Brute-Force PIN Simulation

preset_pin = "0427"
attempts = 0

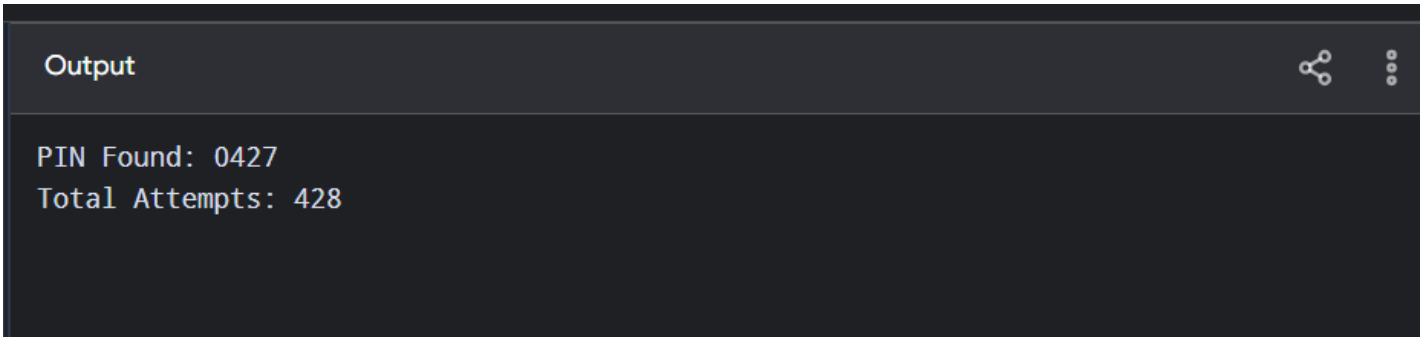
for number in range(10000):
    guess = f"{number:04d}"
    attempts += 1

    if guess == preset_pin:
        print("PIN Found:", guess)
        print("Total Attempts:", attempts)
        break
```

➢ Explanation of the Program

1. **preset_pin:** Stores the correct 4-digit PIN that the program needs to identify.
2. **attempts:** Stores the total number of guesses made by the program.
3. **for loop:** Checks the possible numbers from 0000 to 9999.
4. **f"{number:04d}":** Converts each number into a 4-digit string by adding leading zeros when required.
5. **attempts += 1:** Increases the attempt counter after each guess.
6. **if condition:** Compares the guessed PIN with the preset PIN.
7. **break:** Stops the loop when the correct PIN is found.

➢ **Output**



➢ **Result**

➢ The Python program successfully simulated an offline brute-force process by checking 4-digit PIN combinations. It identified the preset PIN 0427 in 428 attempts and displayed the total number of attempts.

❖ **3. List 4 physical security risks that could affect a server room hosting a popular app like Zomato or BookMyShow, and suggest one mitigation step for each risk.**

➢ **Aim**

- To identify four common physical security risks that could affect a server room hosting an application like Zomato or BookMyShow and suggest suitable mitigation measures for each risk.

➢ **Introduction**

- A **server room** is a dedicated area where servers, networking devices, storage systems, and other IT equipment are installed. Applications such

as Zomato and BookMyShow depend on servers to manage user accounts, orders, bookings, and other important information.

- Physical security is essential to protect server rooms from unauthorized access, environmental problems, and physical damage. If a server room is affected by any physical threat, the application may become unavailable and important data or equipment may be damaged.

➢ Four Physical Security Risks and Their Mitigation

No.	Physical Security Risk	Description	Mitigation Step
1	Unauthorized Physical Access	An unauthorized person may enter the server room and damage equipment, steal hardware, or access sensitive information.	Install biometric access control or smart card systems and allow entry only to authorized employees.
2	Fire Hazard	Electrical faults, overheating equipment, or other causes may lead to a fire that can damage servers and networking equipment.	Install smoke detectors and suitable automatic fire suppression systems, and regularly inspect electrical equipment.
3	Power Failure	A sudden power outage can shut down servers and interrupt services such as online food	Use an Uninterruptible Power Supply (UPS) and backup generators to maintain power during outages.

		ordering or movie ticket booking.	
4	Water Leakage or Flooding	Water from leaking pipes, air-conditioning systems, or flooding may damage servers and other electrical equipment.	Install water leak detection sensors and keep servers away from water pipes and areas where water may collect.

- **Importance of Physical Security**
- Physical security helps organizations in the following ways:
    1. Protects servers and networking equipment from physical damage.
    2. Prevents unauthorized people from accessing important systems.
    3. Reduces the risk of service interruptions.
    4. Protects important business and customer information.
    5. Helps maintain the availability and reliability of online applications.

- **Real-World Example**
- Consider a server room hosting an online food delivery application like Zomato. If a power failure occurs, the servers may stop working and customers may face problems while placing orders. Using a UPS and backup generator helps maintain power and reduces service interruption.
- Similarly, a server room supporting a movie ticket booking application like BookMyShow needs proper access control, fire protection, and water leakage detection to protect its equipment and services.

➢ **Result**

➢ Four physical security risks—Unauthorized Physical Access, Fire Hazard, Power Failure, and Water Leakage or Flooding—were identified. Suitable mitigation measures were suggested for each risk.

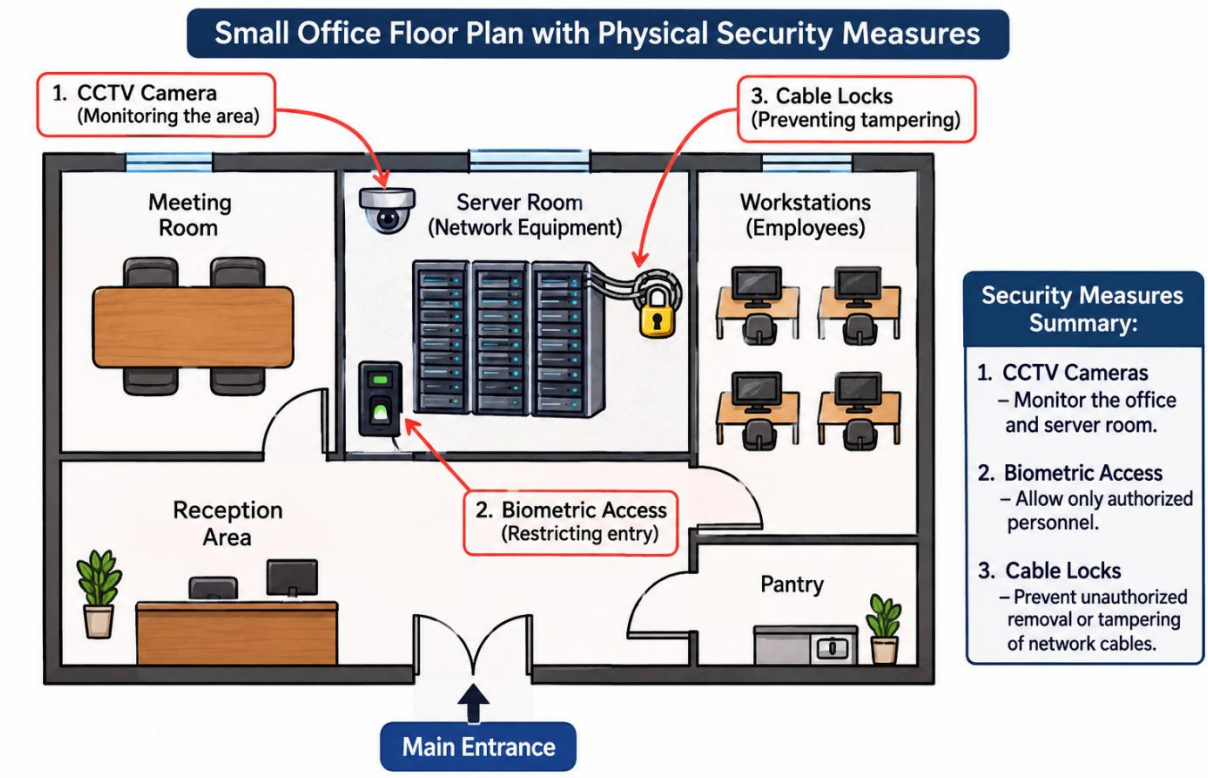
❖ **4. Draw a simple floor plan (hand-drawn or digital) of a small office and mark at least 3 physical security measures you would implement to protect network equipment (such as CCTV, biometric access, or cable locks).**

➢ **Aim**

- To design a simple floor plan of a small office and identify physical security measures to protect network equipment from unauthorized access, theft, and damage.

➢ **Introduction**

- Physical security refers to the protection of computers, servers, networking devices, and other IT equipment from physical threats. A small office should have proper security arrangements to protect its network infrastructure.
- In this floor plan, important areas such as the reception area, meeting room, server room, and employee workstations are included. Three major physical security measures are implemented to improve office security.



- **Physical Security Measures**
  1. **CCTV Cameras:** CCTV cameras are installed near the server room and important office areas to monitor activities and record security events.
  2. **Biometric Access Control:** A biometric access system is installed at the server room entrance. Only authorized employees can enter the room.
  3. **Cable Locks:** Cable locks are used to secure servers and important network equipment, reducing the risk of unauthorized removal or tampering.

➢ **Office Floor Plan Diagram**

➢ The digital floor plan shows the reception area, meeting room, server room, employee workstations, and pantry. It also marks the three physical security measures.

➢ **Result**

➢ A simple office floor plan was designed, and three physical security measures—CCTV cameras, biometric access control, and cable locks—were identified to protect network equipment.

❖ **5. Use ChatGPT or a similar AI tool to generate a checklist of 7 actions a network admin should take to mitigate risks from both network threats and physical security breaches, then review and edit the checklist to make it specific for a college computer lab environment.**

➢ **Aim**

- To generate, review, and customize a checklist of seven security actions that a network administrator should take to protect a college computer lab from network threats and physical security breaches.

➢ **Introduction**

- A college computer lab contains computers, networking devices, servers, printers, and important academic data. These resources may face network threats such as malware, phishing, unauthorized access, and password attacks. Physical risks such as equipment theft, unauthorized entry, and power failure can also affect the lab.

- A network administrator must implement proper security measures to protect the lab's network, devices, and information.

➢ **Seven Security Actions Checklist**

➢ The following checklist was generated using an AI tool and then reviewed and customized specifically for a college computer lab.

No.	Security Action	Implementation in a College Computer Lab
1	Install and Update Antivirus Software	Install reliable antivirus software on all lab computers and regularly update it to detect and prevent malware and other security threats.
2	Use Strong Passwords and Access Control	Provide separate user accounts for students and lab administrators. Use strong passwords and restrict administrator access to authorized staff.
3	Configure Firewall and Network Security	Enable firewalls on lab computers and configure network security rules to block unauthorized connections and unnecessary network traffic.

4	Take Regular Data Backups	Regularly back up important academic files, lab configurations, and server data to a secure storage location to prevent data loss.
5	Install CCTV Cameras and Control Physical Access	Install CCTV cameras near the lab entrance and networking equipment. Allow only authorized staff to access the server or network equipment room.
6	Secure Computers and Networking Equipment	Use cable locks for important devices, keep network cabinets locked, and ensure that computers, routers, and switches cannot be easily removed or tampered with.
7	Maintain Power and Environmental Protection	Use UPS systems for important networking equipment and servers. Ensure proper ventilation, temperature control, and fire safety arrangements in the lab.

- **Review and Customization of the Checklist**
- The initial checklist was reviewed and modified to suit the requirements of a college computer lab.
- The following improvements were made:
  1. Security measures were focused on student computers, faculty systems, and lab servers.
  2. Separate student and administrator accounts were included to prevent unauthorized access.
  3. CCTV cameras and restricted access were included to protect lab equipment.
  4. Regular backups were included to protect academic data and practical assignments.
  5. UPS and proper ventilation were included to reduce the risk of equipment failure.
  6. These changes make the checklist more practical and suitable for a college computer lab.

- **Result**
- A checklist of seven security actions was generated using an AI tool, reviewed, and customized for a college computer lab. It covers network protection, user access control, data backup, physical security, and environmental safety.